

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
1	(a)	(i)		A = theca/thecal cells/follicle cells/granulosa cells (1) B = <u>Primary</u> oocyte/ <u>primary</u> follicle (1) Accept primordial follicle	2			2		1
		(ii)		$8.38 \times 10^{-7} = 3$ marks $83800000 = 2$ marks Accept any correct standard form $83.8 \times 10^{-6} = 2$ marks $4190/0.00005 = 1$ mark Ignore units		3		3	3	
	(b)			Zona pellucida and Corona radiata (1)	1			1		
	(c)			acrosome(1) releases {enzymes/named enzyme} which {hydrolyse/digest/break down} (layers)(1)	2			2		
	(d)	(i)		{ Cortical granules rupture/cortical reaction} + Zona pellucida {thickens/hardens}/fertilisation membrane forms (1) NOT corona radiata thickening (Secondary oocyte taken from graafian follicle is immature) so cannot form a fertilisation membrane/zona pellucida cannot {harden/thicken}/{fewer/less} developed cortical granules (1)	1	1		2		
		(ii)		Any three (x1) from : 1. Fertilisation normally in fallopian tube (1) 2. Embryo needs to reach uterus/ blastocyst to form/allow cleavage to take place(1) 3. If embryo put into uterus immediately, endometrium would not be fully developed/or description (1) 4. {Trophoblastic/chorionic villi} will not have developed (1) 5. {Embryo/blastocyst} would not {implant/survive} (1)			3	3		
				Question 1 total	6	4	3	13	3	1